# Encouraging Vision-Impaired Students to Learn Information Technology Using an e-Learning Platform




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**Abstract** A significant number of visually impaired children are currently studying in Sri Lanka. But in this technological era, this visual impairment is a huge challenge for visually impaired children to get education and to get a good job opportunity. Due to their condition, visually impaired students can't choose a career that fits their preferences and goals. And also, learning IT is more tough for those students because of their visual impairment. In order to overcome this challenge, providing an opportunity to learn through an e-learning platform which includes a broad collection of learning resources that are related to Information Technology. 14 years and above visually impaired students selected as the target audience and this e-learning platform will be specialized based on information technology. Here students can study Python programming, information security, and business management for IT. This e-learning platform will customize the educational experience by suggesting proper modules depending on each learner's requirements and interests using a voice-based chatbot. Emotion detection technology also creates a positive learning environment and offers emotional assistance. Students with vision impairments can learn the fundamentals of programming such as pseudo codes and flow charts with the help of voice commands. Individuals can also use hand gestures for assessments, which increases user friendliness and involvement of the user. This comprehensive project aims to empower visually impaired individuals in Sri Lanka and realize their potential for future success.

**Keywords** Blind · Vision impaired · e-Learning platform · Machine learning

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## 1 Introduction

In the current global environment, the frequency of visual impairments is a major concern, affecting an estimated 191 million people globally, of whom 10.8 million individuals are completely blind [1]. Within this large population, there is an important section of Sri Lanka that faces different challenges. When our research team visited The Ceylon School for the Deaf and Blind (CSDB) and The Employer's Federation of Ceylon, we found out there are difficulties that Sri Lankan visual impairments face especially in the education and work sector, even though they have skills equivalent to their sighted rivals. These difficulties highlight the critical requirement for coordinated initiatives to guarantee fair chances for people with visual impairments, especially in fields such as computer programming, information security, and business management for IT [2, 3].

Research highlights the exceptional learning abilities of the visually impaired population, particularly in technology-related fields [4]. Acknowledging this basic capacity, a customized learning environment has been created to support their interaction with contemporary technology and remove the obstacles preventing them from taking advantage of education and employment opportunities. This research project aims to close the gap between aspiration and reality by providing visually impaired learners with the tools and resources they need to pursue their passions and make valuable contributions to their country.

This study is fundamentally motivated by the belief that every person should have an equal chance to prosper in a world that is becoming more technological, regardless of physical constraints. This application is more than just technological advancements; they are tools of empowerment that provide people the confidence and competence to successfully navigate the complexity of the digital world. Through the integration of an AI chatbot, voice-based flowcharts and algorithms learning, emotion-detecting technology, and interactive assessments using hand gestures, this research aims to improve learning outcomes and encourage comprehensive development in visually impaired individuals. It is imperative to recognize the possible constraints and difficulties that may arise from this undertaking, such as problems with technological accessibility, social complexities, and resource limitations. This research project is an attempt to address the differences in work and education that visually impaired people, especially those in Sri Lanka, encounter. It is evidence of the ability to overcome obstacles and realize human potential through creativity, inclusion, and tenacity. We hope to establish a society in which every person, regardless of social or physical barriers, has the chance to achieve success.

## 2 Literature Review

There is some research that focuses on creating chatbots for e-learning platforms because they can be considered an important innovation in the education field. Teachers face difficulties when it comes to giving attention to each student. As a solution, some studies suggest creating chatbots because they can work with multiple students at a time. Additionally, students can get an interactive learning experience using chatbots [5]. However, these chatbots can't be used by visually impaired students, because they can't use technologies like normal students. As a solution for that problem, we can use voice-based chatbot. Our proposed system presents the realization of a voice-based chatbot for supporting visually impaired students to identify the most suitable module for them and their knowledge level based on the selected module. Additionally, they can use this chatbot to ask questions related to anything in the world [6]. Radolfo, Felippe, and Carlos have researched how to give algorithmic knowledge to vision-impaired children. Theoretical parts like basic algorithm concepts, basic programming, and the theory of flowcharts are taught to vision-impaired children using screen readers. They use screen reading software such as DOSVOX [7], JAWS [8], and NVDA [9] which allow visually impaired students to access text and images on computers. Also, use flowcharts to teach by physically touching their shapes and they read the algorithms by using screen readers. The process in this research paper is commendable. However it can be difficult for a vision-impaired child to do all these things alone, so there should be a teacher or a guide who has knowledge about this. While researching about the vision impaired/blind children of the selected age group, the teachers of CSDB said that if we can teach concepts like basic algorithms, basic programming, and flowcharts interactively to the children even without using a screen reader it would be very valuable. Further, while gathering facts to create an e-learning system suitable for vision-impaired children, the CSDB told us that even though fully blind children learn about flowchart symbols, they do not have much understanding of them, so we should teach them the algorithm concept of flowcharts. Partially blind children should be taught all things like flowcharts and algorithms. There is no need for screen readers due to the theory lessons related to these through a video series so that all vision-impaired individuals can understand them in general. After that, it is possible to listen to the questions of the quizzes given to test the knowledge of vision-impaired children through Text to Speech (TTS).

Johnson et al. (Famuyide and Akindoju [10]) focused on using the Universal Design Model (UDM) as a guide in the creation and assessment of an online voice-assisted examination system (OVAES) [10]. Their study emphasizes how crucial it is to modify technology to meet the unique requirements of visually impaired students. With capabilities such as keyboard navigation and audio support, OVAES showed encouraging results in improving content comprehension and enabling self-assessment tasks. However, when completing the module assessments and responding to questions, our system uses hand gestures.

Sharma et al. explore applying convolutional neural networks (CNNs) to identify facial emotions [11]. The study focuses on the importance of facial expressions in social communication and their importance to interactions between humans and computers. Using the FER 2013 dataset as its training set, the ResNet9 model showed 63% accuracy in identifying different emotions. Furthermore, a GUI desktop program for real-time emotion recognition has been developed, demonstrating the usefulness of the suggested methodology. To address this concern, our system detects six main emotions (Happy, Sad, Angry, Neutral, Surprise, and Fear) to properly identify individuals with visual impairments during the final web assessment through the web camera using CNN and OpenCV2.s.

This research paper compares a learning platform for the visually impaired with existing systems like JAWS, NVDA, and DOSVOX. Traditional platforms focus on screen reading, while the proposed platform combines voice commands with an AI chatbot for more independent interaction between blind and partially impaired users. Our platform offers interactive learning with voice-based flowcharts and algorithms, catering to the cognitive and emotional needs of visually impaired students, bridging the gap between traditional accessibility technologies.

## 3 Methodology

In our proposed e-learning system includes a voice-based chatbot specifically designed to identify the suitable module to learn and the knowledge level of vision-impaired users. Implementing voice-based chatbot includes several steps such as data gathering, machine learning model training, and natural language processing (NLP) techniques.

We have conducted discussion sessions and surveys with teachers and students of the CSDB in order to gain an understanding of the requirements of a voice-based chatbot and to clarify the effectiveness of this method. And also got the help of them to gather datasets and to create questions for stages of chatbot.

When creating intents, divided the responses based on questions. In this step, the most frequent answers that visually impaired students were defined. Then preprocessed the gathered data using methods like tokenization and lemmatization. To increase chatbot's accuracy of output we used NLP techniques including Natural Language Toolkit (NLTK). Tokenization is the process of dividing a text into smaller tokens. Lemmatization is the process of reducing words to their base form.

Several machine learning methods were experimented in order to find out the best model for chatbot. Initially, tested Long Short-Term Memory (LSTM) models. However, we finally selected Artificial Neural Network (ANN) models because of their higher classification accuracy when identifying intents according to user inputs (Fig. 1).

To implement the system, we used speech recognition library to translates voice inputs to text and used Google Text-to-Speech (GTTS) library to deliver the output as voice. As an additional feature linked the ChatGPT API into our chatbot system.

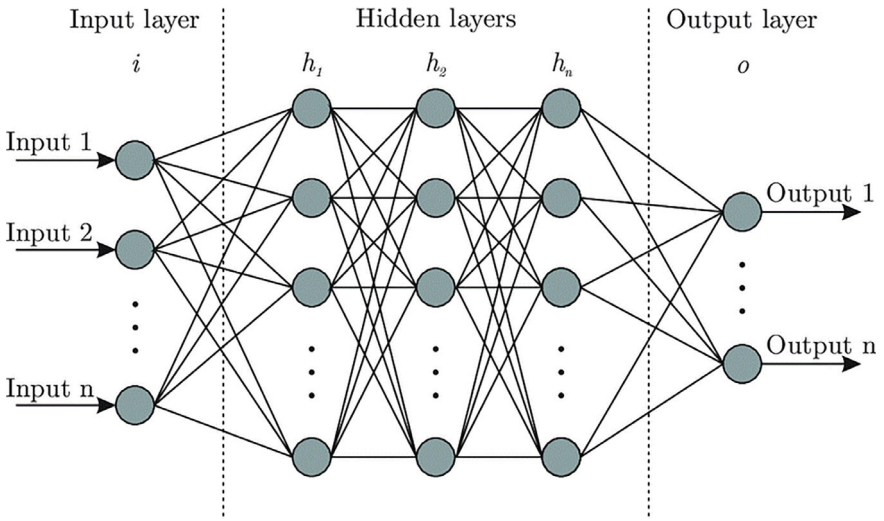



**Fig. 1** Artificial neural network

Because of this integration users can ask any question. There are several significant methods and techniques in providing the proposed solution in the voice selection and answer selection research part. In our web app, when vision-impaired children are given 4 answers to the question, they give the voice command one, two, three, or four as the answer. If the correct answer is 2, when the voice command is given as 2, the SVM machine learning model predicts the corresponding answer. The answer predicted by the model is displayed by STT and the user can listen to the answer given by TTS.

The data set related to this process was created as follows. About 4000–5000 audio clips of people pronouncing one, two, three, and four have been obtained from different voices and the data set needed to train the model has been created. Audacity software has been used as a software tool for this purpose. In addition, there are some special requirements for creating the data set. These audio files should be in WAV format. The sample rate of the audio clip should be 44,100 Hz. Audio recording time duration should be 3 s. And the amplifier/voltage of the audio signal should be between 0.5–1. In the real-time audio recording preprocessing section, the audio clip is cropped with a length of 3 s and low frequencies are cut by basic noise cancelation. The data set is created by this kind of process.

Mel Frequency Cepstral Coefficients (MFCC) are used for data extraction in speech and audio processing tasks. They capture key aspects of human speech while eliminating extraneous information. MFCCs are extracted from Voice Activity Detection speech segments, providing compact and discriminative features for classification models. Support Vector Machine (SVM) is used for classification and regression in high-dimensional feature spaces (Fig. 2).

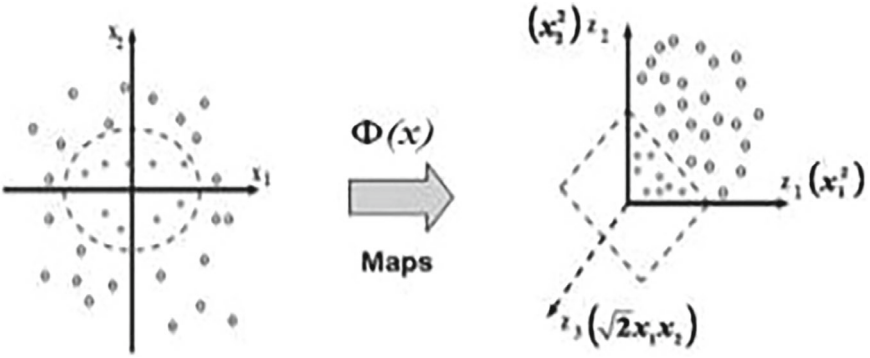



**Fig. 2** Support vector machine

The retrieved MFCC features from the labeled speech samples corresponding to distinct answers (e.g., "answer 1", "answer 2", etc.) would be used to train the SVM classification model in your case. The trained SVM model learns to categories fresh incoming speech segments into one of the predefined classes using the patterns collected by the MFCC features. Before creating the SVM model, two other models, CNN and LSTM, were tried to do this audio processing and classification, but due to the accuracy of the SVM model being relatively higher than the other two and the use of less computational power, it was used for our research (Table 1).

The interactive assessment evaluates and improves visually impaired individual's understanding by utilizing hand gesture technologies. Users can interact with the evaluation tool by using hand gestures to answer questions and get feedback on their replies. Through the use of this approach, the evaluation process will be more inclusive and accessible for individuals with visual impairments, enabling them to have an active role in learning activities pertaining to business management, information security, and basic programming. The evaluation tool gives visually impaired individuals an interesting and creative approach to demonstrate their comprehension of different concepts and topics through the use of hand gesture technology. Moreover, a model was developed, and its architecture was optimized to utilize the minimum computational power effectively after undergoing modifications. The accuracy of the model is 99%. Additionally, the integration of libraries like MediaPipe, OpenCV,

**Table 1** Performance in CNN architecture

| Feature | Model name | Accuracy |
|---|---|---|
| Answer selection model | SVM | 0.99 |
| | CNN | 0.87 |
| | LSTM | 0.64 |
| Voice-based chatbot model | ANN | 0.94 |
| | LSTM | 0.73 |

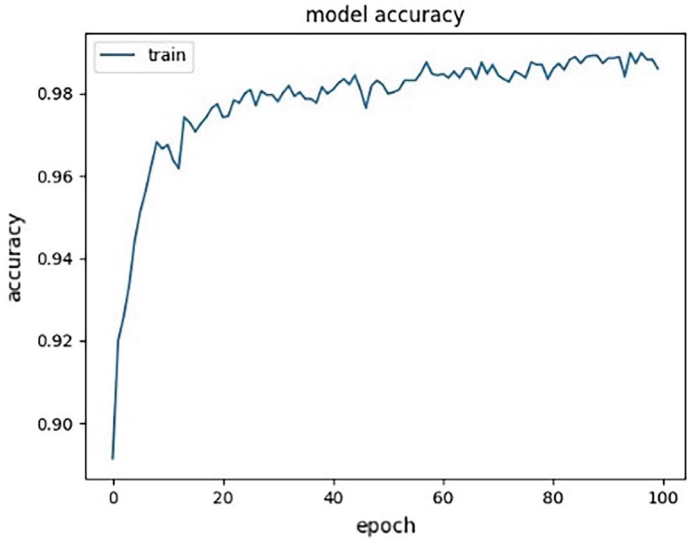



**Fig. 3** Accuracy of hand gesture model

NumPy, Pandas, scikit-learn, and Keras with TensorFlow can enhance the functionality of an assessment tool, improving accuracy and effectiveness. This combination of technologies ensures a robust web-based application infrastructure, benefiting visually impaired individuals and enhancing their learning experience (Fig. 3).

We have invited seven visually impaired members to gather data. We captured their finger movement within 10 min. After that trained the model using that data. We have used a variety of hand gestures and movements to test the robustness of the model across different scenarios.

To support visually challenged individuals during their final online web assessments, this research investigates the design, implementation, and success of a web application coupled with visual technologies such as web camera-based emotion recognition. The study uses a mixed-methods approach to integrate qualitative insights with quantitative data analysis. Depending on the chosen implementation, the React.js and Node.js front-end, supported by a MongoDB database, records both audio (such as pitch and MFCCs) and possibly video data from the web camera [12]. These features are analyzed by deep learning frameworks (TensorFlow, PyTorch, Keras, Pandas), which then extract emotions and categorize them into specific groups (Happy, Sad, Fear, Angry, Neutral, Surprise). These categories are evaluated using a scoring system, which helps to determine the student's general emotional state throughout the test.

Furthermore, the primary focus of this project is to utilize deep learning to categorize emotions, Support Vector Machines (SVM) can be investigated as a backup classification technique in future versions. Additionally, the integration of Natural Language Generation (NLG) could facilitate the provision of customized feedback

to individuals, taking advantage of their assessment results and emotional condition. Potential connections between emotions and assessment performance are investigated using data analysis methods including Item Response Theory (IRT) and Bayesian Knowledge Tracing (BKT). This method seeks to overcome the difficulty of identifying emotions in individuals who are visually challenged by making use of both visual signals and, if used, facial expressions that are recorded by the web camera. In the end, the system aims to give insightful information about individuals' feelings, boost their self-esteem and online web assessment experience, and potentially improve visually impaired individuals' learning experiences in programming, business, and security after O/L and A/L examinations.

The data was annotated with the appropriate emotions. Using TensorFlow or PyTorch, a CNN model was trained to map retrieved features to emotions. Predicting a user's emotional state could be used to customize interactions and information with a voice assistant, possibly providing support or changing the way they communicate. The goal of this project is to improve accessible technology experiences by better understanding emotion detection for people with impaired vision. Each participant contributed a 1.5-min keynote video recording. It consisted of six different emotions (Happy, Sadness, Fear, Neutral, Anger, Surprise). Then a 40-s video recording of the left and right sides, separated by frames, was also contributed (Fig. 4).

Epochs represent the number of times the entire framed emotions images dataset is processed. Positive training and validation accuracy indicate a well-defined model. Preprocessing includes noise reduction, feature extraction, and data enrichment. Python-based deep learning tools like TensorFlow and PyTorch are used to prepare the data. Performance is assessed using measures like accuracy, F1-score, and confusion matrices.

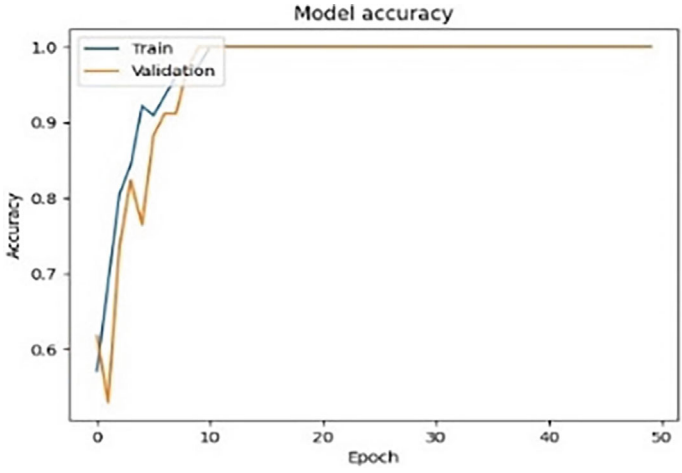



**Fig. 4** Accuracy of emotion detection model

## 4 Result and Discussion

Multiple machine learning and deep learning models were experimented with in the implementation of e-learning platform to find the best methods for different components of the proposed system. Based on accuracy level Convolutional Neural Network (CNN), Support Vector Machine (SVM), and Artificial Neural Network (ANN) are the most effective modules. The selection of models causes to successful implementation of the e-learning platform. Completely blind and partially blind users can use the application with minimal help from guardians and screen readers.

Our study offers a thorough and novel method for meeting the educational needs of people with visual impairments, especially in the area of information technology. The incorporation of voice-activated chatbots, facial recognition software, and hand gesture interactions highlights our dedication to diversity and ease of use. We have included user testing and feedback systems that continuously inform modifications to guarantee the platform's usability and efficacy. To evaluate the viability and efficacy of our solutions in actual educational settings, preliminary pilot studies have been carried out, including testing phases with visually impaired students from the University of Sri Jayewardenepura, Sri Lanka. These studies have yielded important insights into the practical implementation of our solutions. In order to obtain long-term funding and support for the project, possible collaborations with governmental and non-governmental groups are also being investigated. Taking on issues connected to internet connectivity and access to devices among visually impaired individuals in Sri Lanka is also a priority, with strategies in place to mitigate these barriers and ensure broader accessibility (Tables 2, 3 and 4).

This research suggests a voice-based learning platform for visually impaired users, focusing on lighter background colors and larger letter sizes. The platform's user-friendly interface reduces the need for guardians and screen readers, making learning more accessible. The emotion detection technology makes learning more personalized. Interactive learning modules simplify complex IT topics, especially those requiring voice commands and hand gestures. This innovative approach encourages visually impaired students to pursue IT courses and provides self-assurance in using digital technologies. The program shows promise in enabling visually impaired individuals to achieve their academic and professional goals in the IT industry.

## 5 Conclusion and Future Work

The research aims to meet the educational needs of visually impaired individuals in the digital era by creating accessible web apps for Python programming, business management, and information security. Conversational chatbot technology and emotion recognition technology improve learning experiences and foster flexibility and trust. Future research can expand accessibility and empower visually challenged individuals in Sri Lanka by adding language learning features, Braille conversion

**Table 2** Test case for voice-based chatbot

| | |
|---|---|
| Test case ID and title | TC_001<br>Chatbot subject recommendation for visually impaired student |
| Test objective | To verify that the chatbot on the e-learning platform correctly suggests a suitable subject based on the responses provided by the visually impaired student |
| Test steps | 1. Navigate to the interview screen<br>2. Initiate an audio communication with the chatbot<br>3. Answer the questions asked by the chatbot regarding subject preferences:<br>**Ex:**<br>**Question 1**: "What are your interests in terms of subjects?"<br>**Answer**: "I am interested in problem-solving and logical thinking" |
| Expected result | 1. The chatbot accurately processes the responses provided by the visually impaired student<br>2. Based on the responses, the chatbot suggests a suitable subject clearly and audibly |
| Actual result | 1. The chatbot processes the responses accurately and suggests a suitable subject based on the student's answers<br>2. The recommendation is displayed clearly and audibly to the visually impaired student |
| Pass/Fail criteria | **Pass**: The chatbot accurately processes the responses and provides a suitable subject recommendation |

**Table 3** Test case for voice commands and answer selection

| | |
|---|---|
| Test case ID and title | TC_002<br>Quiz navigation and answering questions by voice command |
| Test objective | To verify that the visually impaired student can effectively navigate through the quiz section, hear the questions, and provide answers using voice commands |
| Test steps | 1. The quiz section is accessible and contains questions related to the subject<br>2. The quiz questions and answer choices are designed to be audible<br>3. The student listens to the questions and answers |
| Expected result | 1. The student effectively communicates with the quiz interface using voice commands<br>2. The system accurately recognizes and selects the answer choices based on the voice commands |
| Actual result | 1. The student successfully communicates with the quiz interface using voice commands<br>2. The system accurately recognizes and selects the answer choices based on the voice commands |
| Pass/Fail criteria | **Pass**: The visually impaired student successfully navigates through the quiz section, effectively communicates with the quiz interface using voice commands, and completes the quiz without encountering technical issues |

**Table 4** Test case for hand gestures

| | |
|---|---|
| Test case ID and title | TC_003<br>Quiz interaction via hand gestures for visually impaired student |
| Test objective | To verify that the visually impaired student can effectively interact with the quiz section using hand gestures to select answer choices |
| Test steps | 1. Ensure that the hand gesture recognition feature is enabled and functional<br>2. Listen to the quiz and use hand gestures to select the answer (Ex: raise one finger to select answer choice 1, two fingers for choice 2, and so on.)<br>3. Verify that the system accurately recognizes and selects the answer choices based on the hand gestures |
| Expected result | 1. The student effectively uses hand gestures to interact with the quiz interface for selecting answer choices<br>2. The system accurately recognizes and selects the answer choices based on the hand gestures |
| Actual result | 1. The student successfully uses hand gestures to interact with the quiz interface for selecting answer choices<br>2. The system accurately recognizes and selects the answer choices based on the hand gestures |
| Pass/Fail criteria | **Pass**: The visually impaired student successfully interacts with the quiz section using hand gestures, accurately selects answer choices, and completes the quiz without encountering technical issues |

tools, enhanced evaluations, customized learning pathways, and a community forum for peer-to-peer encouragement. Through the process of this research, we expect to expand accessibility and empower visually challenged individuals in Sri Lanka more than ever. In the future, researchers can add a feature to learn the content using other languages. It will help to serve a wider audience. And also, can add Braille conversion tools to convert the module content and learning material so that users can experience their familiar method of learning. To improve user experience, can create enhanced evaluations and customized learning pathways. And last, a community forum will encourage peer-to-peer encouragement and support. These new paths aim to transform the process of obtaining IT skills into something accessible and strong.

## References

1. Khairallah M, Kahloun R, Bourne R, Limburg H, Flaxman SR (2015) Number of people blind or visually impaired by cataract worldwide and in world regions, 1990 to 2010. Investig Ophthalmol Vis Sci 11
2. Besnoy K, Sandra M, Frances K (2005) Screening students with visual impairments for intellectual giftedness: a pilot study. Rehabil Educ Blind Vis Impairment 7
3. Wickramaarachchi C, Jayathilaka R, Suraweera T, Thelijjagoda S (2023) Can visual impairment impact your income potential? PLoS ONE 23

4. Pires AC, Rocha F, de Barros Neto AJ, Simao H, Nicolau H (2020) Exploring accessible programming with educators and visually impaired children. Assoc Comput Mach 13
5. Colace F, De Santo M, Lombardi M, Pascale F, Pietrosanto A (2018) Chatbot for E-learning: a case of study. Int J Mech Eng Robot Res 7:6
6. Pereira RM, Silla C, Silva F (2018) Teaching algorithms for visually impaired and blind students using physical. Institute of Electrical and Electronics Engineers
7. Barbosa LC, Marques de Sá LAC (2018) MAPVOICE: computational tool to aid in learning cartography for the visually impaired. Bull Geod Sci 11
8. Sarwar U, Eika E (2019) Visual reading vs. screen reading
9. Rosa AD, Justice D (2016) WebReader: a screen reader for everyone, everywhere, p 2
10. Famuyide JA, Akindoju O (2023) Visos: an assistive computer based testing (CBT) in the examination of the visually impaired in Nigeria
11. Jaiswal A, Raju K, Deb S (2020) Facial emotion detection using deep learning. Institute of Electrical and Electronics Engineers
12. Paulose S, Mathew D, Thomas A (2017) Performance evaluation of different modeling methods and classifiers with MFCC and IHC features for speaker recognition. In: Conference on advances in computing and communications